A sequence pinned by its own partial sums

$$S_n = n^2 f(n) \text{ forces } f(n) = \frac{2f(1)}{n(n+1)}$$

Problem. A function f on the positive integers satisfies

$$\sum_{k=1}^{n} f(k) = n^2 f(n) \qquad \text{for every } n \geq 1, \qquad f(1) = 2027.$$

Find $f(2026)$.

$$\boxed{f(n) = \frac{2f(1)}{n(n+1)}, \qquad f(2026) = \frac{2 \cdot 2027}{2026 \cdot 2027} = \frac{1}{1013}.}$$

Overview. A condition relating a sequence to its own partial sums is really a first-order recurrence in disguise. Subtracting the relation at $n-1$ from the relation at n eliminates the sum entirely and leaves

$$(n+1)f(n) = (n-1)f(n-1),$$

whose solution is a telescoping product collapsing to $\frac{2}{n(n+1)}$ times $f(1)$. The value 2027 then cancels against the 2027 hidden inside $2026 \cdot 2027$, which is why the answer is a clean $\frac{1}{1013}$.

1. From the sum condition to a recurrence

Write

$$S_n = \sum_{k=1}^{n} f(k), \qquad \text{so the hypothesis is} \qquad S_n = n^2 f(n) \quad (n \geq 1). \tag{1}$$

For $n = 1$ this reads $f(1) = 1 \cdot f(1)$, which is automatic: the hypothesis places no constraint on $f(1)$, and that is why the problem must supply it.

For $n \geq 2$, subtract (1) at $n-1$ from (1) at n, using $S_n - S_{n-1} = f(n)$:

$$n^2 f(n) - (n-1)^2 f(n-1) = f(n).$$

Collecting the $f(n)$ terms,

$$(n^2 - 1)f(n) = (n-1)^2 f(n-1),$$

and factoring $n^2 - 1 = (n-1)(n+1)$, then cancelling the common factor $n - 1 \neq 0$,

$$\boxed{(n+1)f(n) = (n-1)f(n-1), \qquad n \geq 2.} \tag{2}$$

Equivalently

$$\frac{f(n)}{f(n-1)} = \frac{n-1}{n+1}, \qquad n \geq 2. \tag{3}$$

At $n = 2$ relation (2) reads $3f(2) = 1 \cdot f(1)$, so $f(2) = \frac{f(1)}{3}$; the recurrence is well posed from the start, and the sum has disappeared completely.

2. Solving the recurrence

Iterating (3) from n down to 2,

$$f(n) = f(1)\prod_{k=2}^{n}\frac{k-1}{k+1} = f(1)\cdot\frac{1\cdot 2\cdot 3\cdots(n-1)}{3\cdot 4\cdot 5\cdots(n+1)}. \tag{4}$$

The numerator is $(n-1)!$ and the denominator is $\frac{(n+1)!}{2}$, so the product collapses:

$$\prod_{k=2}^{n}\frac{k-1}{k+1} = \frac{(n-1)!}{(n+1)!/2} = \frac{2}{n(n+1)}.$$

Hence

$$\boxed{f(n) = \frac{2f(1)}{n(n+1)}, \qquad n \geq 1,} \tag{5}$$

and the formula is also correct at $n = 1$, where it returns $\frac{2f(1)}{2} = f(1)$.

The mechanism is a shifted telescope: the numerator of each factor is the denominator of the factor two steps earlier, so all but two terms cancel at each end, one pair surviving at the bottom $(1\cdot 2)$ and one pair at the top $(n(n+1))$.

3. Verification

Formula (5) must still be checked against (1), not merely against (2), since the passage to the recurrence discards information. With $f(k) = \frac{2f(1)}{k(k+1)}$,

$$S_n = 2f(1)\sum_{k=1}^{n}\frac{1}{k(k+1)} = 2f(1)\sum_{k=1}^{n}\left(\frac{1}{k} - \frac{1}{k+1}\right) = 2f(1)\left(1 - \frac{1}{n+1}\right) = \frac{2f(1)\,n}{n+1},$$

while

$$n^2 f(n) = n^2\cdot\frac{2f(1)}{n(n+1)} = \frac{2f(1)\,n}{n+1}.$$

The two agree for all $n \geq 1$, so (5) is the unique solution. $\square$

The partial-fraction telescope $\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}$ is the same collapse as in Section 2, now on the additive side, a sign that n^2 was not an arbitrary weight.

4. The answer

With $f(1) = 2027$ and $n = 2026$,

$$f(2026) = \frac{2\cdot 2027}{2026\cdot 2027} = \frac{2}{2026} = \frac{1}{1013}.$$

The constant 2027 cancels identically: it appears in the numerator through $f(1)$ and in the denominator as the factor $n+1 = 2027$. The problem is arranged so that the requested index is exactly one below the given initial value, which is the only reason the answer is rational with such a small denominator. Since $2026 = 2\cdot 1013$, the final form is $\frac{1}{1013}$.

n	$f(n) = \frac{2\cdot 2027}{n(n+1)}$	S_n	$n^2 f(n)$
1	2027	2027	2027
2	2027/3	4054/3	4054/3
3	2027/6	$3 \cdot 2027/2$	$3 \cdot 2027/2$
4	2027/10	$8 \cdot 2027/5$	$8 \cdot 2027/5$

5. The general weight

Nothing is special about n^2 except the arithmetic that follows. Suppose instead

$$\sum_{k=1}^{n} f(k) = w(n) f(n), \qquad n \geq 1,$$

for a given sequence w. Consistency at $n = 1$ forces $w(1) = 1$. The same subtraction gives

$$\bigl(w(n) - 1\bigr) f(n) = w(n-1) f(n-1), \qquad \text{so} \qquad f(n) = f(1) \prod_{j=2}^{n} \frac{w(j-1)}{w(j)-1}, \tag{6}$$

valid whenever $w(n) \neq 1$ for $n \geq 2$.

$w(n)$	ratio $\dfrac{w(n-1)}{w(n)-1}$	solution
n	$\dfrac{n-1}{n-1} = 1$	$f(n) = f(1)$: the condition $S_n = nf(n)$ says every term equals the running average, i.e. f is constant.
n^2	$\dfrac{(n-1)^2}{n^2-1} = \dfrac{n-1}{n+1}$	$f(n) = \dfrac{2f(1)}{n(n+1)}$, the present problem.
$2n-1$	$\dfrac{2n-3}{2n-2}$	$f(n) = f(1) \prod_{j=2}^{n} \dfrac{2j-3}{2j-2} = f(1) \dfrac{(2n-3)!!}{(2n-2)!!} \sim \dfrac{Cf(1)}{\sqrt{n}}$.
n^3	$\dfrac{(n-1)^2}{n^2+n+1}$	no elementary closed form; the denominator n^2+n+1 does not factor over $\mathbb{Q}$, so the product does not telescope.

The table isolates what makes the stated problem work: for $w(n) = n^2$ the difference $w(n) - 1 = n^2 - 1$ factors, and one of its factors cancels the $(n-1)^2$ coming from $w(n-1)$. A weight is "nice" for this problem exactly when $w(n) - 1$ shares a factor with $w(n-1)$, and $n^2 - 1 = (n-1)(n+1)$ is the smallest interesting instance.

6. A continuous shadow

The same computation has an integral analogue that explains the exponent. If $F(x) = \int_1^x f$ satisfies $F(x) = x^2 f(x) = x^2 F'(x)$, then

$$\frac{F'}{F} = \frac{1}{x^2}, \qquad \log F(x) = -\frac{1}{x} + C, \qquad F(x) = A\,\mathrm{e}^{-1/x},$$

so $f(x) = F'(x) = \frac{A}{x^2}\mathrm{e}^{-1/x}$, which decays like x^{-2}. The discrete solution $f(n) = \frac{2f(1)}{n(n+1)}$ has the same n^{-2} decay; the discrete problem simply replaces the exponential factor by an exact rational one, because the discrete telescope is exact where the continuous one requires an integrating factor.